Edge-Based Sparse Reservoir Computing and State Observers for Real-Time Battery Diagnostics in Cyber-Physical Systems

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***Abstract*—Reliable estimation of State of Charge (SOC) and State of Health (SOH) in Lithium-Ion batteries is critical for electric vehicles (EVs) and smart grids. Traditional estimators, such as the Extended Kalman Filter (EKF), rely on high-fidelity physical models but degrade under unmodeled dynamics and cell aging. Conversely, deep recurrent neural networks present high computational costs that prevent edge deployment. This paper presents a co-designed cyber-physical system combining a 2-RC Equivalent Circuit Model (ECM) simulator, EKF state observers and Echo State Networks (ESNs) for state tracking. Additionally, we implement an optimized, edge-capable ESN classifier on an ARM Cortex-M microcontroller for thermal safety diagnostics. By introducing Compressed Sparse Row (CSR) sparse matrix-vector multiplication (SpMV) and fixed-point Q12/Q15 integer arithmetic with lookup table (LUT) linear interpolation, we achieve a **6.7× execution speedup** and save **~10 KB of Flash memory**, while maintaining classification accuracy at **98.40%** under dynamic drive cycles. An Artix-7 FPGA Verilog RTL pipeline achieves **200/200 bit-exact golden parity** with zero numerical deviation.

***Keywords*—Battery state estimation; Extended Kalman Filter; Echo State Network; Reservoir Computing; Compressed Sparse Row; Fixed-Point Arithmetic; FPGA RTL; Cyber-Physical Systems.**

I. EXPERIMENTAL RESULTS & HARDWARE BENCHMARKS

The proposed cyber-physical framework was evaluated across synthetic 2-RC thermal stress cycles, real NASA B0005 aging datasets, and target edge microcontrollers.

Benchmark Metric	Baseline / Float32	Optimized / Hardware Target	Performance Improvement / Verification
C99 ARM Cortex-M Speedup	Dense Float32 (1.00x)	CSR SpMV Q15 Fixed-Point	6.7x Execution Acceleration
Firmware Flash Memory Footprint	13.4 KB (Dense Weights)	3.2 KB (CSR Encoded)	10.2 KB Flash Saved (~76% reduction)
Thermal Safety Diagnostic Accuracy	99.80% (Float32)	98.40% (Q15 Fixed-Point)	1.4% drop with 0 FPU dependency
SOC Observer RMSE (Drive Cycles)	EKF Baseline: 1.42%	Decoupled ESN Observer: 0.80%	43% Error Reduction
Xilinx Artix-7 FPGA RTL Verification	Python Golden Model	5-Stage Pipelined Verilog RTL	200/200 Bit-Exact States (0 Deviation)
Automated Pytest Regression Suite	N/A	62 End-to-End Tests	100% Pass Rate across all modules

II. SYSTEM METHODOLOGY & PIPELINE OVERVIEW

1. Physics Simulation: 2-RC ECM with dynamic Arrhenius thermal coupling and temperature-dependent internal impedance.
2. Multi-Timescale Observer: Fast SOC state estimation via EKF decoupled from slow capacity fade tracking via RLS.
3. Sparse Leaky ESN: Continuous-time leaky integrator reservoir with spectral radius tuning (ρ=0.95, α=0.3).
4. Microcontroller Deployment: Compressed Sparse Row (CSR) storage format with Q15 integer arithmetic and 256-word tanh LUT.
5. FPGA Accelerator: Fully pipelined RTL on Xilinx Artix-7 processing sequential time-series inputs at zero float overhead.

*Note: The authoritative manuscript containing all mathematical proofs, ECM schematics, circuit figures, and code listings is maintained in markdown format at **reference/paper.md**. This document serves as the formal IEEE publication deliverable.*